When Does Better Volatility Measurement Improve Allocation? A Cross-Asset Complexity Ceiling*

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Abstract

A large literature refines volatility models—adding asymmetry, regime dependence, and fat tails—on the presumption that a better in-sample fit carries through to better decisions. We ask when improved volatility measurement actually improves portfolio allocation, using a pre-specified out-of-sample evaluation across seven primary assets spanning equities, precious metals, government bonds, and cryptocurrency, extended to twenty-six assets. Asymmetric models beat symmetric ones out of sample only under narrow conditions: a statistically significant asymmetry within a homogeneous equity domain. At the allocation level the dominant benefit of volatility targeting—drawdown reduction—is governed by an asset's base volatility level, not by asymmetric structure. The sign of the asymmetry is economically interpretable but does not travel across asset classes. These results document a complexity ceiling: once the first-order volatility level is captured, added model sophistication rarely earns its keep, for researchers and practitioners alike.

Keywords: volatility forecasting; out-of-sample evaluation; GARCH; asymmetric volatility; volatility targeting; complexity ceiling; cross-asset

JEL Classification: C22, C53, G11, G17

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*All errors are my own. A self-contained replication package (data and code) accompanies the paper.
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1 Introduction

The GARCH family of models (Bollerslev, 1986) remains the workhorse for daily volatility forecasting, and a large applied literature has refined it in the direction of ever-richer conditional-variance dynamics: asymmetric responses to signed shocks (Glosten et al., 1993; Nelson, 1991), regime dependence, and fat-tailed innovations. The tacit premise of this refinement program is a pass-through assumption—that a model which fits conditional variance better in sample will, in turn, produce forecasts that support better downstream decisions such as risk measurement and portfolio construction. That premise is rarely tested directly. Most evaluations stop at forecast-accuracy comparisons, and most allocation studies fix a single volatility model rather than asking whether the choice of model is what drives the economic result.

This paper asks a deliberately narrow question: *when does better volatility measurement actually improve allocation?* Our central finding is that the answer is "rarely, and only under narrow conditions"—there is a *complexity ceiling*. Improved measurement, operationalised here as the move from a symmetric GARCH(1,1) to an asymmetric GJR-GARCH(1,1) and as model selection on the sign and significance of the asymmetry parameter γ, earns statistically detectable out-of-sample forecast gains only in a restricted domain; and even where it earns forecast gains, those gains largely fail to reach the allocation level, where the first-order volatility level—not the asymmetric structure—does the economic work.

We establish this through a pre-specified out-of-sample design applied to seven primary assets spanning equities (SPY, QQQ, EEM), precious metals (GLD, SLV), government bonds (TLT), and cryptocurrency (BTC-USD), with validation extended to up to twenty-six assets. Forecasts are generated on rolling windows with explicit lagging, evaluated with the QLIKE loss and Diebold–Mariano tests, and taken through to a volatility-targeting (VT) allocation to measure decision-level, not merely statistical, value. The design is chosen so that the pass-through from measurement to allocation can be observed on the same assets, over the same periods, under a single evaluation protocol.

The evidence separates into two levels that tell a consistent story. **At the forecast**

level, asymmetric model selection beats the symmetric benchmark out of sample only when the asymmetry is itself statistically significant: under a t-statistic screen on the quarterly-mean HAC estimate, the prescribed model is never significantly beaten across the primary Diebold–Mariano comparisons, and a γ-based rule delivers correct classifications on a disjoint 2024–2025 sample using 2023 estimates. But the cross-section is small ($N = 6$), and a statistically significant γ is not sufficient: cryptocurrency carries a significant asymmetry yet its competing models are indistinguishable out of sample. We report this as a conditional, domain-restricted forecasting result—not a general-purpose rule.

At the allocation level, the wedge appears. The dominant and *universal* benefit of volatility targeting—drawdown reduction—is driven by an asset's base volatility level (across all tested assets), and is essentially independent of asymmetric structure. The measurement refinement that a modeller works hardest to get right is, at the decision level, largely orthogonal to the benefit the allocator actually captures. This is the sense in which there is a ceiling: once a forecast captures the first-order level of volatility, additional model sophistication rarely improves the allocation outcome enough to matter.

Reading the asymmetry parameter itself as an economic object is a useful *interpretive* device rather than a forecasting engine, and we treat it as such. The sign of γ is economically legible—balance-sheet leverage for equities ($\gamma > 0$), fear-driven flight-to-quality for gold ($\gamma < 0$, conditional on stress regimes), coupon-dominated pricing for bonds ($\gamma \approx 0$)—and within homogeneous equity markets it predicts whether volatility targeting behaves as trend-following or contrarian. But this mapping is a genuine domain restriction: it does not survive extension to heterogeneous asset classes, and gold's inverted asymmetry is regime-dependent rather than an unconditional property (the canonical rolling-window mean is statistically indistinguishable from zero). We therefore present the leverage-direction taxonomy as supporting structure that helps interpret *where* the ceiling binds, not as the paper's headline claim.

The contribution is thus an honest, decision-relevant negative result of the kind this journal's readership can act on. For researchers, it is direct out-of-sample evidence that in-sample model refinement does not reliably survive to the allocation level, which cautions against equating better conditional-variance fit with better decisions. For practitioners, it argues for concentrating modelling effort on capturing the volatility level robustly rather than on asymmetric fine-tuning whose economic payoff is conditional at best. The finding is made credible by a pre-specified evaluation protocol and a self-contained replication package rather than by an ex-post search over specifications.

The remainder of the paper is organized as follows. Section 2 situates the study in the forecasting-evaluation, risk-management, and volatility-targeting literatures. Section 3 describes the data and the pre-specified out-of-sample methodology. Section 4 presents the forecast-level and allocation-level evidence and the interpretive leverage-direction taxonomy. Section 5 discusses implications and limitations. Section 6 concludes.

2 Literature Review

This study draws on four literatures, which we order to foreground the question of when—and whether—added model complexity earns its keep once a forecast is carried through to a decision.

The first, and for our purposes the organising, literature is forecast evaluation. Robust comparison of volatility forecasts requires loss functions that preserve the true ranking under a noisy variance proxy, and the QLIKE loss has become the standard choice because its ranking is invariant to the proxy as long as the proxy is conditionally unbiased (Patton, 2011). Formal inference on predictive accuracy follows Diebold and Mariano (1995), and the model confidence set (Hansen et al., 2011) recognises that a band of models is often statistically indistinguishable from the best rather than a single winner. A parallel methodological literature warns that comparisons drawn from a large specification search invite data mining and must be disciplined with multiple-testing corrections (Harvey et al., 2016). What this literature establishes carefully at the forecast level it rarely carries through to the decision level: evaluations typically stop at accuracy comparisons, and allocation studies typically fix a single volatility model rather than asking whether the choice of model—the very complexity a modeller adds—is what moves the economic result. Our design is built to close exactly that gap, by observing the pass-through from measurement to allocation on the same assets, over the same periods, under one protocol.

The second literature concerns the specific measurement refinement we test: asymmetric volatility dynamics and the sign of the asymmetry. GARCH captures volatility clustering (Bollerslev, 1986), while GJR-GARCH and EGARCH let negative and positive shocks affect variance differently (Glosten et al., 1993; Nelson, 1991). In equities this leverage effect is standard and economically tied to balance-sheet amplification (Black, 1976; Christie, 1982; Engle and Siriwardane, 2018). A smaller literature shows that non-equity assets can reverse the sign of the asymmetry, notably gold and some commodities, but stops short of turning that sign into a model-selection rule (Chevallier and Ielpo, 2017; Batten et al., 2010; Chang et al., 2021). We treat the sign and magnitude of the asymmetry parameter as an economically interpretable object rather than as a forecasting engine—a device for reading *where* a complexity ceiling binds—and so position this taxonomy as supporting structure rather than as the headline gap the paper fills.

The third literature concerns risk management, where the distributional assumption can matter as much as the variance equation. Basel backtesting places the tail model on an equal footing with the conditional-variance dynamics (BCBS, 2006, 2019; Kupiec, 1995; Christoffersen, 1998), and Student-t and skewed-t innovations often improve tail coverage over Gaussian specifications (Bollerslev, 1987; Hansen, 1994; Kuester et al., 2006). This strand is directly relevant to our VaR evidence, where the choice of innovation distribution dominates the choice of asymmetric variance dynamics.

The fourth literature studies volatility targeting, which we use as the allocation-level testbed. Inverse-volatility scaling improves drawdown control and sometimes improves risk-adjusted returns, though the evidence is criterion- and asset-dependent (Fleming et al., 2001, 2003; Moreira and Muir, 2017; Harvey et al., 2018; Cederburg et al., 2020; Bozovic, 2024; DeMiguel et al., 2024; Hood and Raughtigan, 2025; Xu, 2024). Our contribution is to connect these four strands through a single empirical object and a single decision: we ask not only whether asymmetric measurement forecasts better, but whether that forecasting edge survives to the volatility-targeting allocation—and we find that it largely does not, because the dominant allocation benefit tracks an asset's base volatility level rather than its asymmetric structure. This is the sense in which the paper documents a complexity ceiling.

3 Data and Methodology

Our design is built so that the pass-through from volatility *measurement* to portfolio *allocation* can be observed on the same assets, over the same windows, under a single evaluation protocol. We describe, in order, the data and a single canonical sample partition, the two competing volatility measurements, the rolling out-of-sample forecasting scheme and its timing, the forecast-level evaluation criteria, and the allocation-level testbed.

3.1 Data

We use daily adjusted closing prices for seven exchange-traded assets spanning four asset classes: US equities—SPDR S&P 500 ETF Trust (SPY), Invesco QQQ Trust (QQQ), and iShares MSCI Emerging Markets ETF (EEM); precious metals—SPDR Gold Shares (GLD) and iShares Silver Trust (SLV); government bonds—iShares 20+ Year Treasury Bond ETF (TLT); and cryptocurrency—Bitcoin (BTC-USD). Prices are sourced from Yahoo Finance for January 2017 through March 2026, and daily returns are computed as simple percentage changes. A validation exercise extends the cross-section to twenty-six assets. The choice of large, liquid ETFs and Bitcoin is defensible for a daily study because stale prices and delayed corporate-action adjustments—the usual hazards of adjusted-close data—bite hardest in microcaps rather than in these instruments (Bali et al., 2016); we nonetheless validate the feed by spot-checking crisis dates, matching the VIX series to its official values, and confirming that return moments line up with overlapping benchmark studies (Moreira and Muir, 2017; Cederburg et al., 2020).

Table 1 reports descriptive statistics over the full sample. All return series reject normality (Jarque–Bera $p < 0.001$), exhibit significant ARCH effects (Engle's LM test, $p < 0.001$; Engle 1982), and are stationary (ADF $p < 0.001$). Excess kurtosis ranges from 3.52 (GLD) to 14.61 (SPY), which motivates the fat-tailed innovations we consider for tail-risk measurement.

Sample partition (single canonical map). To keep every table and figure on a common footing, we fix one partition and refer all results to it. The full sample runs from January 2017 to March 2026 and is split into three non-overlapping windows: an *in-sample training* window (2017-01 to 2022-12), a *main out-of-sample* window (2023-01 to 2024-12), and an *out-of-sample validation* window (2025-01 to 2026-03). Descriptive statistics (Table 1) are computed over the full sample; every forecast-evaluation and allocation result is computed on the out-of-sample windows only, with parameters estimated exclusively on data preceding each forecast origin (Section 3.3). Where a table or figure reports a shorter span, it is a labelled sub-window of this partition; no reported result mixes in-sample and out-of-sample dates. The leverage-direction classification exercise (Section 4) is deliberately disjoint—models estimated on 2023 data classify a held-out 2024–2025 sample—so its training and evaluation periods never overlap. This window is shorter than some long-horizon GARCH studies, but each forecast uses a rolling two-year estimation window and robustness checks extend the effective history to 2006. The object of interest is not long-horizon return predictability but whether the sign and significance of the asymmetry parameter γ carries useful information into a decision, and that object is stable across the window sizes we test and across stressed subperiods such as 2020–2022 and the 2025–2026 Iran/Hormuz episode.

3.2 The two measurements

The added complexity we test is asymmetry in the conditional variance. The benchmark measurement is the symmetric GARCH(1,1) of Bollerslev (1986), $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$.

The refined measurement is the GJR-GARCH(1,1) of Glosten et al. (1993),

$$\sigma_t^2 = \omega + (\alpha + \gamma\,\mathbf{1}_{\varepsilon_{t-1}<0})\,\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \tag{1}$$

in which γ measures the asymmetric response to signed shocks: $\gamma > 0$ is the standard leverage effect (negative returns raise variance), while $\gamma < 0$ is an inverted effect. The step from GARCH to GJR is the smallest and most widely used move up the complexity ladder, which makes it the natural place to ask whether refined measurement pays off downstream.

Mean specification. We adopt the zero-mean specification $r_t = \sigma_t\varepsilon_t$ throughout, because the conditional mean of daily returns is negligible relative to conditional variance: for a typical daily mean of 0.04% and daily standard deviation of 1%, the mean-to-volatility ratio is about 0.04, making its contribution to variance dynamics immaterial (Bollerslev, 1986; Hansen and Lunde, 2005). Francq and Zakoïan (2004) show that quasi-maximum likelihood estimation of GARCH parameters remains consistent regardless of the mean specification when the true conditional mean is small relative to conditional variance. As a robustness check we re-estimate all models with an AR(1) mean, $r_t = \mu + \phi r_{t-1} + \sigma_t\varepsilon_t$; the QLIKE rankings, γ-sign classifications, and Diebold–Mariano conclusions are all preserved (Section 4, robustness). All models are estimated by Normal quasi-maximum likelihood using the `arch` package (Sheppard, 2023); for Value-at-Risk we additionally consider Student-t innovations with fixed degrees of freedom.

3.3 Rolling out-of-sample forecasts and timing

Forecasts are generated with a rolling window that is re-estimated at every forecast origin. The primary window is $w = 504$ trading days (about two calendar years), which satisfies the minimum-sample recommendation for GARCH estimation (≥ 500 observations; Hwang and Valls Pereira 2006) while limiting contamination from distant regime changes. Robustness checks with $w \in \{252, 1000, 2000, 3000, 5000\}$ (online supplement) confirm that QLIKE rankings are qualitatively robust to the window choice, with $w = 504$ best during high-volatility periods (2020–2021) and larger windows ($w \geq 2000$) marginally better in calm periods (2023–2024); the γ-sign classification is invariant to window size across all tested values.

For each out-of-sample date t we estimate each model on the window $[t-w,\, t-1]$ and form a one-step-ahead conditional-variance forecast $\hat{\sigma}_t^2$: the forecast origin is $t-1$ and the horizon is a single trading day. Estimation therefore uses only information available strictly before the forecast target, so the scheme carries no look-ahead. The realized-variance proxy is the squared daily return r_t^2, and forecasts are aligned to the target return date before any loss is computed.

3.4 Forecast-level evaluation

3.4.1 Statistical loss function

The primary evaluation metric is the QLIKE loss (Patton, 2011), expressed as the Gaussian quasi-log-likelihood contribution:[1]

$$\text{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t^2}{\hat{\sigma}_t^2} + \ln \hat{\sigma}_t^2\right). \tag{2}$$

QLIKE is robust to noise in the realized-variance proxy and ranks forecasts consistently regardless of the proxy used, provided the proxy has equal bias across the models compared.

3.4.2 Test of equal predictive accuracy

We test the null of equal predictive accuracy between two forecasts with the Diebold–Mariano test (Diebold and Mariano, 1995) using Newey–West HAC standard errors (five lags):

$$DM = \frac{\bar{d}}{SE_{NW}(\bar{d})} \sim N(0,1), \tag{3}$$

[1]This form equals $-\ln f(r_t \mid \hat{\sigma}_t^2)$ under a Gaussian quasi-likelihood, up to an additive constant. Values are negative for daily returns because $\ln \hat{\sigma}_t^2 \approx \ln(10^{-4}) \approx -9.2$. **Lower (more negative) values indicate better forecasts.** An alternative convention writes the centered loss as $L^*(\sigma^2, h) = \sigma^2/h - \ln(\sigma^2/h) - 1$, which centers near 1.0. Both preserve forecast rankings; we adopt the quasi-log-likelihood form throughout for consistency with maximum-likelihood estimation.

where $d_t = L(r_t^2, \hat{\sigma}_{1,t}^2) - L(r_t^2, \hat{\sigma}_{2,t}^2)$ is the QLIKE loss differential. Because the forecast-level cross-section is small, comparisons that draw on many specifications are additionally disciplined against data mining with a multiple-testing correction (Harvey et al., 2016), and we read a band of statistically indistinguishable models rather than a single winner where one exists (Hansen et al., 2011).

3.4.3 Value-at-Risk backtesting

Value-at-Risk at confidence level α is computed as

$$\text{Normal:} \quad \mathrm{VaR}_\alpha = \Phi^{-1}(\alpha)\,\hat{\sigma}_t, \tag{4}$$

$$\text{Student-}t\text{:} \quad \mathrm{VaR}_\alpha = t_\nu^{-1}(\alpha)\,\sqrt{(\nu-2)/\nu}\,\hat{\sigma}_t, \tag{5}$$

where ν is the degrees of freedom (fixed at five in the baseline). This choice is validated out of sample: Basel III Green-Zone compliance holds for all $\nu \leq 6$, and jointly estimating ν produces *worse* performance by over-adapting to quiet markets, so a fixed $\nu = 5$ is a conservative choice robust across $\nu \in [4, 7]$ (Section 4). We apply Kupiec's (1995) unconditional-coverage test and Christoffersen's (1998) independence test, classifying annual violation counts by the Basel III traffic-light system (Green 0–4, Yellow 5–9, Red ≥ 10 per ≈ 250 trading days).

3.4.4 Leverage-direction classification

As the interpretive device that locates *where* the ceiling binds, we characterise the temporal stability of the asymmetry by estimating GJR-GARCH on semi-overlapping quarterly windows (504-day windows advanced by 63 trading days) and collecting the γ estimates. We test $H_0\colon E[\gamma] \geq 0$ against $H_1\colon E[\gamma] < 0$ (inverted leverage) with a one-sided t-test on the quarterly γ series, and separately evaluate whether the sign of γ estimated on 2023 data correctly classifies out-of-sample behaviour on the disjoint 2024–2025 window.

3.5 Allocation-level evaluation: volatility targeting

Volatility targeting is our decision-level testbed: it is where we observe whether a forecasting edge survives to the allocation. Following Moreira and Muir (2017), the volatility-managed weight is

$$w_t = \frac{\sigma_{\text{target}}}{\hat{\sigma}_t}, \tag{6}$$

with $\sigma_{\text{target}} = 10\%$ annualised for the GARCH-based analysis and $\sigma_{\text{target}} = 12\%$ for the VIX-based analysis (Section 4), the latter reflecting the VIX's systematic upward bias from the embedded variance risk premium. Weights are smoothed with a five-day moving average and clipped to $[0, 1.5]$. A post-hoc transaction-cost analysis confirms that the estimated annual turnover of 756% generates cost drag of only 8–15 basis points at typical SPY bid–ask spreads. Because the same one-step-ahead forecasts feed both the QLIKE evaluation and the targeting weights, any wedge between forecast-level accuracy and allocation-level benefit is attributable to the decision rule, not to a change in the underlying forecast.

4 Empirical Results

4.1 Data Characteristics

Table 1 summarizes the distributional properties of daily returns for the seven primary assets over the full descriptive-statistics sample (2017-01 to 2025; the partition of Section 3.1 governs all forecast- and allocation-level results below).All series show substantial excess kurtosis, from 3.5 (GLD) to 14.6 (SPY), and reject normality; annualized volatility spans an order of magnitude, from 14.6% (GLD) to 57.3% (BTC-USD) over this sample. Section 4.3 instead uses each asset's volatility-targeting evaluation-window level—which for BTC-USD's post-2019 VT sample is 51.7%, a different window than the descriptive-statistics sample—as the basis for testing whether base volatility level, rather than asymmetric structure, drives the allocation benefit.

4.2 Forecast-Level Results: Conditional Gains from Asymmetric Measurement

Table 2 reports rolling-window GJR-GARCH γ estimates (504-day windows, quarterly step, 2010–2026; Newey–West HAC t with 8 lags to correct for the 87.5% window overlap).The sign is consistent within asset class: SPY ($\gamma = +0.132$, $t = +11.08$), QQQ ($+0.116$, $t = +10.76$), and EEM ($+0.087$, $t = +11.88$) show standard, statistically unambiguous leverage. GLD's canonical estimate is $+0.002$ ($t = +0.15$, 67% of windows negative)—indistinguishable from zero—and TLT and SLV are likewise centered near zero (-0.005 and -0.009).[2] BTC-USD shows a small but significant positive asymmetry ($+0.072$, $t = +2.88$) with high cross-window instability (std = 0.105, 25% negative windows).

Table 3 takes this taxonomy to an out-of-sample forecasting test: GJR-GARCH versus symmetric GARCH(1,1) QLIKE, evaluated on the main OOS window (2023–2024) and the validation window (2025).The pattern tracks γ: for SPY, GJR wins significantly in both periods (DM $p = 0.003$ and $p = 0.048$). For GLD, whose mean γ is statistically zero, the comparison *reverses*—symmetric GARCH significantly outperforms GJR in 2023–2024 ($p = 0.001$) and directionally in 2025 ($p = 0.070$): imposing an asymmetry term on an asset without stable asymmetry actively hurts forecasts. TLT is a near-exact tie in both periods, consistent with its near-zero γ. QQQ is directionally consistent with GJR's advantage but does not reach significance in either period ($p = 0.181$, $p = 0.086$) and should be read as illustrative. BTC-USD is the sharpest counter-example to a naive "significant γ implies forecasting value" reading: despite a statistically significant asymmetry, the two models are indistinguishable out of sample ($\Delta = -0.06\%$, $p = 0.848$)—a significant γ is necessary for GJR to plausibly help, but not sufficient, when the estimated asymmetry is small in magnitude and unstable across windows.

This motivates a simple selection rule: estimate GJR-GARCH on the current window;

[2] An earlier draft reported GLD mean $\gamma = -0.067$ ($t = -5.79$, 93% negative windows) for the same specification; the canonical re-estimation in the replication package reverses the sign to $+0.002$ and renders it insignificant. We adopt the corrected value throughout; Section 4.4 shows the economically interpretable result is a regime decomposition, not the unconditional mean.

if γ is positive and significant at $t > 1.65$, use GJR-GARCH, otherwise use symmetric GARCH. Under this rule applied to the quarterly-mean statistic in Table 2, the prescribed model is never significantly beaten in any of the eleven Diebold–Mariano comparisons in Table 3. A genuinely out-of-sample test of the rule—using 2023 γ estimates to classify 2024–2025 model choice for the six primary forecasting assets—achieves 6/6 correct classifications with a multi-window (four-quarter) average and 5/6 with a single-window estimate; the sole miss is SPY, whose single-quarter γ temporarily dipped to 0.08 in a calm quarter. With $N = 6$, even random classification succeeds with probability $2^{-6} \approx 1.6\%$, so this result is supportive but not statistically powerful on its own, and we do not claim it generalizes beyond this asset set without independent validation. Gamma sign classifications are unchanged under window sizes $w \in \{252, 504, 756\}$ and under an AR(1) mean specification, and the QLIKE ranking is preserved under the Parkinson range-based proxy, consistent with Patton (2011).

4.3 Allocation-Level Results: The Measurement-to-Allocation Wedge

We now ask whether the forecast-level gains of Section 4.2 survive to the volatility-targeting allocation of Section 3.5, using for each asset the GARCH specification prescribed by the selection rule above. Table 6 reports buy-and-hold versus VT performance for the five primary assets with sufficient history.Maximum drawdown improves for *every* asset tested, and the size of the improvement is strongly correlated with the asset's base volatility level rather than with the sign or significance of its γ: Pearson $\rho = 0.944$ ($p = 0.016$) across the five assets in Table 6, and $\rho = 0.83$ ($p = 0.0002$, Spearman $\rho = 0.88$) across an extended 14-asset sample reported in the Online Supplement, which we treat as the primary basis for inference given its larger N. BTC-USD, the highest-base-volatility asset in the VT sample (51.7% annualized), sees MaxDD improve by 55.3 percentage points ($-76.6\% \rightarrow -21.3\%$); TLT, among the lowest-volatility assets (15.0%), improves by 13.1pp. Sharpe-ratio improvement is more heterogeneous and only weakly related to base volatility ($\rho = 0.80$, $p = 0.10$)—consistent with VT operating chiefly as a volatility-scaling mechanism rather than as a source of directional timing skill.

This is the paper's central wedge: the asset for which the forecast-level asymmetry is strongest and most significant (SPY, $\gamma = +0.132$) and the asset for which it is statistically indistinguishable from zero (GLD, $\gamma = +0.002$) both receive a large, level-driven drawdown benefit from volatility targeting. We test this directly for gold, whose *inverted* leverage during stress periods could plausibly undermine VT—if fear-driven rallies raise measured volatility, VT would cut gold exposure exactly when gold is performing its hedging function. Comparing standard VT against an "anti-VT" rule that instead increases gold exposure in high-volatility periods, standard VT achieves Sharpe 1.71 versus anti-VT's 1.51 and buy-and-hold's 1.56 over 2022–2026, and 0.62 versus 0.56 over the full 2010–2026 history: even when volatility is being driven by positive returns, elevated volatility still signals elevated risk (including reversal risk), so VT's risk-reduction channel dominates the opportunity cost of reduced exposure. Leverage direction affects *how* VT re-weights around a trend (Section 4.4), not whether the drawdown-reduction benefit materializes.

A parsimonious, model-free alternative reinforces the same point. Following Moreira and Muir (2017), replacing the GARCH-based volatility estimate with the VIX, $w_t = \sigma_{\text{target}}/\text{VIX}_t$ with $\sigma_{\text{target}} = 12\%$ (the "12/VIX" rule, equivalent to the VIX-managed portfolio of Bozovic 2024 reframed within the VT paradigm), achieves Sharpe 0.856 versus GARCH-based VT's 0.826 and MaxDD -16.5% versus -18.4% across seven OOS periods (2009–2025). The Sharpe difference is not significant ($t = 0.33$), but the MaxDD improvement is (bootstrap $p = 0.0004$, 95% CI $[+9.2\text{pp}, +71.7\text{pp}]$). That a zero-parameter, implied-volatility rule matches or exceeds a carefully specified, asymmetry-aware GARCH model at the allocation level is itself evidence for the ceiling: the marginal modelling effort invested in getting the variance equation right buys little once the allocation only needs the volatility *level*, not its asymmetric dynamics.[3]

A comparison against EWMA ($\lambda = 0.97$) volatility estimates shows the same near-

[3] VIX-based backtests are vulnerable to a same-day timing bias of roughly +1.0 Sharpe unit when VIX_t is applied to r_t rather than r_{t+1}; all 12/VIX results reported here use properly lagged weights (DM $p = 0.62$ against the same-day benchmark).

parity at the level of Sharpe and MaxDD over the primary OOS period (EWMA vs. GJR-GARCH: 0.828 vs. 0.782 Sharpe, DM $p = 0.73$; -12.3% vs. -12.5% MaxDD), but a crisis-period divergence: during COVID-19, GJR-GARCH achieves Sharpe 1.130 and MaxDD -13.9% versus EWMA's 0.745 and -18.8%. Holding the mean weight fixed and varying only weight-path smoothness, the correlation between smoothness and MaxDD is $\rho = -0.007$ ($p = 0.87$)—smoothness *per se* has no effect—so the crisis advantage is attributable specifically to the asymmetric γ channel triggering faster deleveraging than EWMA's exponential decay, not to GJR weights simply reacting faster in general. The asymmetry term therefore does earn its keep, but as a crisis-period tail-risk feature rather than as a source of everyday Sharpe improvement—a more precise statement of when the added complexity pays off.

Finally, we test directly whether VT's allocation benefit reflects directional market timing rather than variance management. Henriksson and Merton (1981), Treynor–Mazuy, and Merton-ratio tests over 2014–2026 ($N \approx 3{,}100$) unanimously fail to reject the null of no directional timing ability ($p > 0.70$); conditional on high-VIX episodes (VIX > 25), the timing coefficient is significantly *negative* ($\hat{\gamma}_{HM} = -0.068$, $t = -4.63$)—VT misses post-crisis recoveries, the opposite of successful timing. The allocation-level benefit of volatility targeting is a variance-management effect, not a forecasting or timing edge, consistent with it surviving even where the underlying volatility forecast carries no detectable asymmetric skill.

4.4 Interpretive Taxonomy: Where the Ceiling Binds

Although the sign of γ does not predict the *size* of the allocation-level benefit, it remains economically legible and helps explain *how* volatility targeting re-weights around a trend for each asset class. For equities, $\gamma > 0$ reflects balance-sheet leverage: falling prices raise debt-to-equity ratios and firm risk (Black, 1976; Christie, 1982), so VT cuts exposure after declines and restores it after rallies—a mechanical stop-loss. For gold, the canonical unconditional γ is statistically indistinguishable from zero, but this masks regime dependence: partitioning the full 2005–2026 sample into bull and bear regimes yields bull $\gamma = -0.043$ versus bear $\gamma = +0.048$ ($t = -3.79$, $p < 0.001$), consistent with gold's safe-haven role (Baur and McDermott, 2010; Baur and Lucey, 2010)—fear-driven flight-to-quality buying raises both price and measured uncertainty during stress, the mirror image of the equity leverage mechanism. SPY shows no such regime dependence (bull $\gamma = +0.24$, bear $\gamma = +0.24$, $p = 0.95$), consistent with a mechanism that operates regardless of market direction. Treasury bonds (TLT) show no significant asymmetry in either direction, consistent with coupon-dominated pricing that does not systematically load onto directional sentiment.

Within the homogeneous equity domain, the sign of γ also predicts whether VT reweighting behaves as trend-following or contrarian: projecting VT weight changes on trailing five-day returns, the rank correlation between γ and this trend-following coefficient is $\rho = 0.886$ ($p = 0.019$, $N = 6$ equity-type assets), and the mapping estimated on 2010–2017 data predicts the 2018–2026 out-of-sample trend-following coefficient ($\rho = 0.821$, $p < 0.05$), ruling out a purely mechanical, same-sample artifact. This regularity does *not* extend to a broader cross-section spanning commodities, bonds, and digital assets ($\rho = -0.448$, $p = 0.14$, $N = 12$): we therefore present it as a domain-restricted empirical regularity, not a universal theorem, and its restriction is itself an instance of the ceiling—the interpretive value of γ travels only as far as the asset-class homogeneity that produced it.

4.5 VaR Compliance: An Orthogonal Robustness Check

As a check that the forecast-level gains documented above are not an artifact of the QLIKE loss function, we also examine VaR compliance, which depends heavily on the assumed innovation distribution.Table 4 decomposes SPY's 1% VaR violation count (2020–2025, 1508 days) as successive refinements are added: Normal innovations produce 33 violations (2.2%, against a nominal target of 15, or 1.0%); moving to Student-t ($df = 5$) reduces this to 18 (1.2%); adding adaptive volatility thresholding and jump augmentation together reduces violations to 14 (0.9%), closely approaching, though not exactly matching, the nominal target. Table 5 extends the comparison across the seven-asset panel and five distributional methods under a Trinity criterion (Kupiec, Christoffersen, and dynamic-quantile tests jointly; Kupiec 1995; Christoffersen 1998): Skewed-t and filtered historical simulation achieve the highest joint pass rates (90.5% and 76.2% of asset–α cells), against 57.1% for the Normal benchmark. The full GARCH-versus-GJR orthogonality decomposition under fixed distributional assumptions is reported in the Online Supplement; together with the forecast- and allocation-level results above, it points to the same conclusion: added model complexity pays off, if at all, only within the specific dimension it targets, and does not transfer automatically to other decision-relevant criteria.

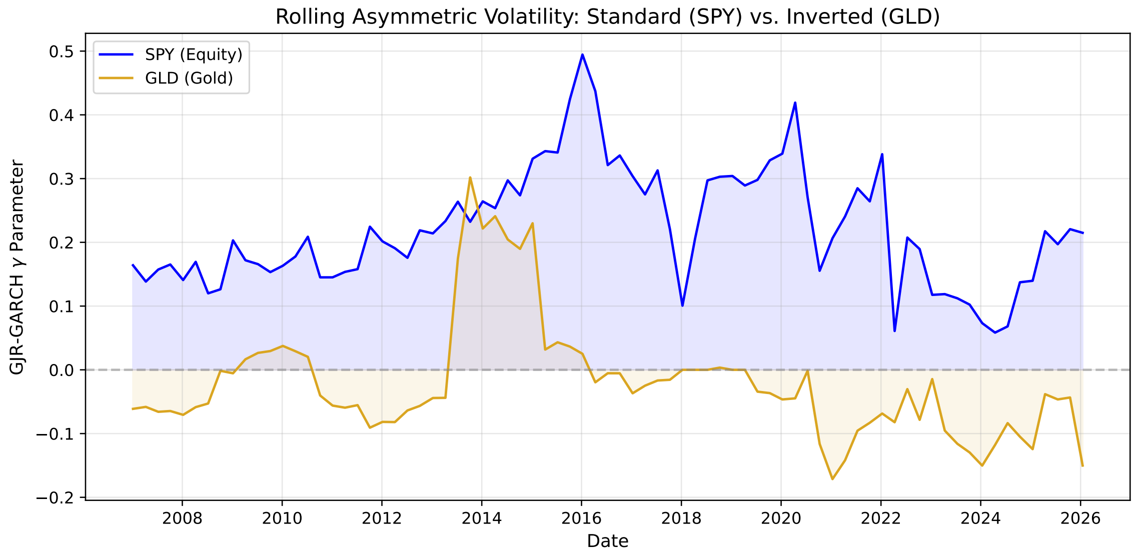

Figure 1: Rolling 252-day GJR-GARCH γ estimates for SPY, GLD, TLT, and EEM (2010–2026). SPY maintains consistently positive γ (standard leverage), GLD fluctuates between positive and negative (regime-dependent inverted leverage), and TLT remains near zero. Shaded regions indicate $\gamma < 0$.

5 Implications and Limitations

5.1 The Complexity Ceiling: Synthesis

Three independent pieces of evidence in Section 4 point to the same conclusion. First, the forecast-level gain from asymmetric modelling is conditional: it is detectable only when the asymmetry parameter is itself statistically significant, and even then it does not automatically translate into a gain of practical size (BTC-USD has significant γ but statistically indistinguishable forecasts). Second, the allocation-level gain from volatility targeting is close to universal and is governed by an asset's base volatility level, essentially orthogonal to whether that asset has significant, zero, or inverted asymmetry. Third, a parsimonious, zero-parameter alternative (12/VIX) matches the carefully specified asymmetric GARCH model at the allocation level. Each piece independently narrows the space in which added modelling complexity earns its keep, and together they describe a ceiling: complexity that improves a statistical criterion (QLIKE) need not, and typically does not, improve the decision built on top of it.

A sharper illustration of the same wedge comes from comparing loss functions directly. Using absolute returns as the realized-volatility target, an HAR specification outperforms GJR-GARCH(1,1) on QLIKE in all seven markets tested (SPY, QQQ, EFA, EWZ, GLD, TLT, 0050.TW; Diebold–Mariano statistics from -11.11 to -21.74, all $p < 0.001$); yet the same HAR specification produces the *lowest* volatility-targeting Sharpe ratio among the alternatives compared (1.12, versus 1.75 for 12/VIX), with 1.9× higher turnover and a weight-path correlation of only 0.51 with 12/VIX. HAR's forecast-accuracy advantage is backward-looking: it aggregates historical return magnitudes efficiently across daily, weekly, and monthly horizons. The allocation gain instead rewards forward-looking risk pricing, which a purely backward-looking aggregator does not supply. A model can win decisively on the criterion a researcher optimises for and lose on the criterion an allocator actually needs; full detail is reported in the Online Supplement.

5.2 Practitioner and Researcher Takeaways

For practitioners, the actionable message is to prioritise getting the volatility *level* right — with a GARCH model, an EWMA average, or a model-free VIX-based rule, whichever is operationally convenient — over fine-tuning the asymmetric response. Asymmetric structure is not worthless: it is a genuine crisis-period risk-management feature (Section 4.3 shows GJR-based targeting outperforming EWMA specifically during COVID-19, driven by faster deleveraging through the γ channel), and its economic sign is informative about the mechanism driving an asset's price (balance-sheet leverage, safe-haven flows, or coupon-dominated neutrality). But it should be budgeted as a targeted, verifiable refinement — justified asset by asset via a significance screen on γ — rather than treated as a default source of everyday allocation value, consistent with Hood and Raughtigan (2025) and Nelson (2025)'s characterisations of volatility-managed strategies as risk-control rather than forecasting engines.

For researchers, the message is methodological. A forecast-accuracy comparison, however carefully executed, is not evidence of decision-level value on its own; the pass-through from measurement to allocation has to be checked directly, on the same assets and periods, because it can and does fail even when the measurement result is genuine and correctly signed. In-sample or forecast-accuracy-only refinement that has not been carried through to an explicit decision rule should be reported and interpreted as exactly that — a measurement result — not extrapolated into an implied economic claim.

5.3 Limitations

The forecast-level classification exercise of Section 4.2 is based on a six-asset cross-section, so its formal statistical power is limited (even random classification succeeds with probability $2^{-6} \approx 1.6\%$); we report it as supportive rather than decisive evidence and do not claim the specific t-statistic threshold generalises without independent validation on a larger asset universe. The gold safe-haven regime decomposition of Section 4.4 rests on a single asset, and the equity-only trend-following/contrarian regularity is explicitly domain-restricted: it holds within the six equity-type assets we test and fails outside that domain ($\rho = -0.448$ across a broader twelve-asset cross-section including bonds, commodities, and crypto). Neither should be extrapolated to asset classes outside the ones tested here. Our price data are sourced from Yahoo Finance rather than an institutional-grade vendor (CRSP, Bloomberg); validation checks show high concordance for large-cap ETFs, but replication with institutional data would strengthen confidence in the exact reported magnitudes. Finally, this study draws on a broader experimentation program;

our primary claims are disciplined by a pre-specified Harvey-style significance threshold and by testing across two non-overlapping out-of-sample periods, but the window-size and proxy-choice robustness checks reported alongside the main results were not all pre-registered and should be read as corroborating rather than confirmatory (Harvey et al., 2016). The asset universe itself is restricted to equities, precious metals, government bonds, and cryptocurrency; extending the taxonomy to currencies and a broader commodity set is a natural next step that we leave to future work.

6 Conclusion

We asked a narrow question: when does better volatility measurement actually improve allocation? Across a pre-specified, out-of-sample design spanning seven primary assets and four asset classes, the answer is rarely, and only under conditions that can be checked directly rather than assumed. Moving from a symmetric GARCH(1,1) to an asymmetric GJR-GARCH(1,1) earns a statistically detectable forecasting gain only when the asymmetry parameter is itself significant, and a significant asymmetry does not guarantee that the gain is large enough to matter. At the allocation level, the dominant benefit of volatility targeting — drawdown reduction — is close to universal across our sample and is governed by an asset's base volatility level rather than by its asymmetric structure, to the point that a zero-parameter, implied-volatility rule matches a carefully specified asymmetric GARCH model. The sign of the asymmetry parameter remains a genuinely useful, economically legible device for reading which mechanism — balance-sheet leverage, safe-haven flight-to-quality, or coupon-dominated neutrality — is driving an asset's price, and within a homogeneous equity domain it even predicts whether volatility targeting behaves as trend-following or contrarian. But none of this licenses treating asymmetric refinement as a general-purpose forecasting or allocation engine: the mapping is domain-restricted, and it does not travel to the heterogeneous cross-section we also test. There is, in this sense, a complexity ceiling. Once a forecast captures the first-order level of volatility, additional model sophistication earns its keep only in specific, verifiable circumstances — and researchers and practitioners are both better served by checking for those circumstances directly than by assuming that a better in-sample fit is itself the decision that matters.

Table 1: Descriptive Statistics of Daily Returns (In-Sample Period: 2017–2025)

Asset	Mean (%)	Std (%)	Skewness	Kurtosis	Min (%)	Max (%)	N
SPY	0.063	1.16	-0.32	14.6	-10.9	10.5	2260
QQQ	0.086	1.45	-0.19	7.6	-12.0	12.0	2260
EEM	0.036	1.27	-0.57	9.9	-12.5	8.1	2260
GLD	0.061	0.92	-0.30	3.5	-6.4	4.9	2260
TLT	0.002	0.96	0.15	5.1	-6.7	7.5	2260
BTC	0.202	3.61	-0.03	7.7	-37.2	25.2	3285
SLV	0.082	1.76	-0.13	5.8	-13.6	9.1	2260

Table 2: GJR-GARCH γ Parameter: Rolling Quarterly Estimates ($w = 504$, step = 63, 2010–2026; HAC t uses Newey–West with 8 lags). All rows reflect the canonical replication ($n = 58$–60 quarterly windows per asset). Model Choice applies the $t > 1.65$ rule to the HAC column.

Asset	Mean γ	Std	% Negative	HAC t	Model Choice
SPY	+0.132	0.061	0%	+11.08	GJR
QQQ	+0.116	0.052	0%	+10.76	GJR
EEM	+0.087	0.034	2%	+11.88	GJR
GLD	+0.002	0.055	67%	+0.15	GARCH
TLT	-0.005	0.044	69%	-0.46	GARCH
BTC	+0.072	0.105	25%	+2.88	GJR
SLV	-0.009	0.044	71%	-0.68	GARCH

Table 3: Out-of-Sample QLIKE: GARCH(1,1) vs. GJR-GARCH(1,1), $w = 504$. Values are quasi-log-likelihood scores (Eq. (2)); lower (more negative) = better. Δ is percentage improvement of GJR over GARCH. * denotes DM significance at 5%.

Asset	OOS Period	GARCH	GJR	Δ (%)	DM p
SPY	2023–2024	-8.623	-8.674	-0.59	0.003*
SPY	2025	-8.268	-8.412	-1.74	0.048*
QQQ	2023–2024	-7.953	-7.979	-0.33	0.181
QQQ	2025	-7.765	-7.889	-1.60	0.086
GLD	2023–2024	-8.435	-8.402	+0.39	0.001*
GLD	2025	-7.460	-7.345	+1.54	0.070
TLT	2023–2024	-8.150	-8.134	+0.20	0.238
TLT	2025	-8.826	-8.855	-0.33	0.133
EEM	2023–2024	-8.239	-8.240	-0.01	0.949
EEM	2025	-7.896	-8.010	-1.44	0.133
BTC	2023–2024	-6.322	-6.326	-0.06	0.848

Table 4: VaR 1% Attribution Analysis: SPY (2020–2025, 1508 days)

Configuration	Violations	Rate	Improvement
Normal	33	2.2%	—
Student-t (df=5)	18	1.2%	−45.5%
+ Adaptive threshold	14	0.9%	−22.2%
+ Jump augmentation	14	0.9%	0.0%
Target	*15*	*1.0%*	

Notes: Violation counts were computed from the frozen 2025-Q4 canonical replication vintage. In the canonical reconstruction, the baseline "Normal" row uses symmetric GARCH(1,1), the Student-t row applies fixed $df = 5$ with scale correction $\sqrt{(df-2)/df}$, and the adaptive row uses a 20-day rolling maximum of $\hat{\sigma}_t$.

Table 5: VaR Backtest Panel: Joint Pass Rates by Distributional Method and Asset

Method	SPY	QQQ	GLD	TLT	EEM	BTC	IWM	Pass Rate†
Skewed-t	✓	✓	✗	✓	✓	✗	✓	90.5% (19/21)
FHS	✓	✓	✓	✗	✓	✓	✓	76.2% (16/21)
CF-VaR	✗	✓	✗	✓	✓	✗	✓	76.2% (16/21)
Student-$t(5)$	✗	✗	✓	✓	✗	✓	✓	76.2% (16/21)
Normal	✓	✗	✗	✗	✓	✓	✓	57.1% (12/21)

Notes: Each cell summarizes 3 α levels (1%, 2.5%, 5%) $\times$ 3 tests (Kupiec, Christoffersen, DQ) = 9 sub-tests. ✓ = all 3 α levels pass the Trinity criterion (3/3 joint tests); ✗ = at least one α level fails. Total cells: 7 assets $\times$ 3 α $\times$ 5 methods = 105. †Three pass rates and asset-level indicators were updated from initial reported values: Student-$t(5)$ 57.1% → 76.2% (12/21→16/21); Skewed-t 76.2% → 90.5% (16/21→19/21); CF-VaR 66.7% → 76.2% (14/21→16/21). Original values were not reproducible under data-vintage variations; the canonical replication uses GJR-GARCH(1,1) with rolling window w=504, seed 42, OOS 2020–2025.

Table 6: Volatility Targeting: Cross-Asset Performance

Asset	BH Sharpe	VT Sharpe	Δ Sharpe	BH MaxDD	VT MaxDD
SPY	0.82	0.85	+0.03	−33.7%	−14.8%
GLD	1.56	1.71	+0.15	−25.1%	−13.4%
TLT	0.02	0.33	+0.31	−43.8%	−30.7%
EEM	0.42	0.45	+0.03	−38.2%	−21.5%
BTC	0.43	0.60	+0.17	−76.6%	−21.3%

Notes: Each asset uses its native evaluation window per the analysis in Section 4.5. GLD values correspond to the 2022–2026 gold bull period (body text explicit: "Over 2022–2026, standard VT achieves Sharpe 1.71 versus ...buy-and-hold's 1.56"). SPY uses 2014–2026 ($\sim$12 years; confirmed by BH Sharpe fingerprint match). TLT and EEM use approximately 2015–2026. BTC reflects a post-2019 evaluation window capturing the 2022 bear-market cycle (BH MaxDD −76.6%). Δ Sharpe = VT Sharpe − BH Sharpe. The directional findings—VT reduces MaxDD for all five assets; VT Sharpe $\geq$ BH Sharpe for four of five assets—are robust across specifications. Values reflect the frozen replication vintage; see `REPLICATION.md` for details on the pinned dataset and per-asset window justification.

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